

Basic Electronics

Tutorial 3

1. In the circuit of Fig. 1, determine the current labeled i with the assistance of nodal analysis techniques.

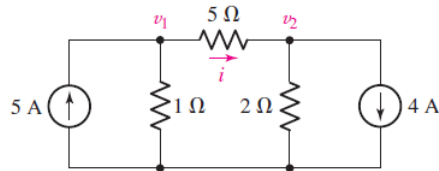


Figure 1: Prob.1

2. Use nodal analysis to find v_P in the circuit shown in Fig. 2

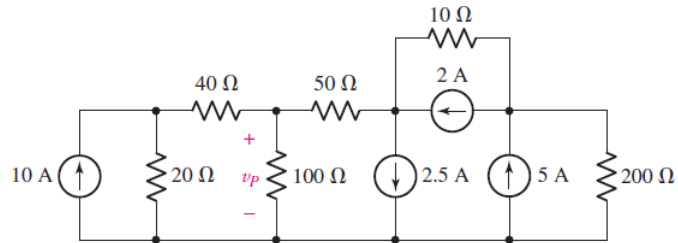


Figure 2: Prob.2

3. For the circuit of Fig. 3, use nodal analysis to determine the current i_5 .

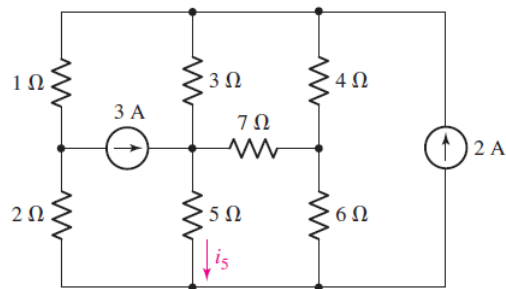


Figure 3: Prob.3

4. Determine the current i_2 as labeled in the circuit of Fig. 4, with the assistance of nodal analysis.

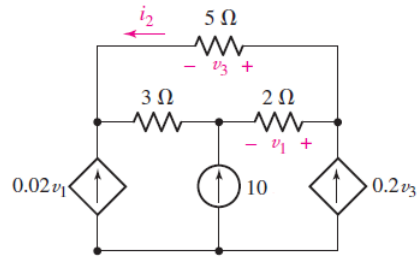


Figure 4: Prob.4

5. Determine the nodal voltages as labeled in Fig. 5, making use of the supernode technique as appropriate.

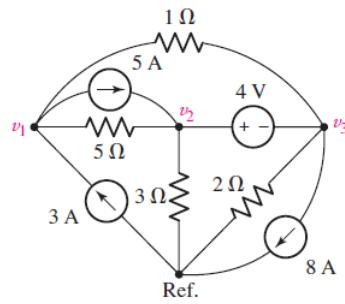


Figure 5: Prob.5

6. Determine the voltage v_x in the circuit of Fig. 6, and the power supplied by the 1 A source.

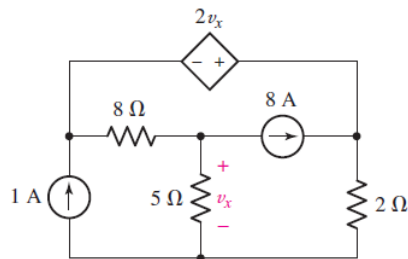


Figure 6: Prob.6

7. For the circuit of Fig. 7, determine all four nodal voltages.

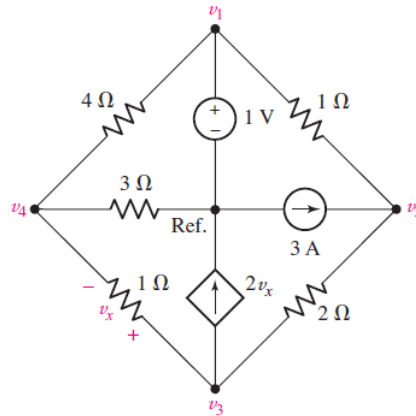


Figure 7: Prob.7

8. Use mesh analysis as appropriate to determine the two mesh currents labeled in Fig. 8

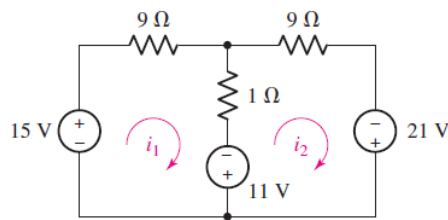


Figure 8: Prob.8

9. Choose nonzero values for the three voltage sources of Fig. 9 so that no current flows through any resistor in the circuit.

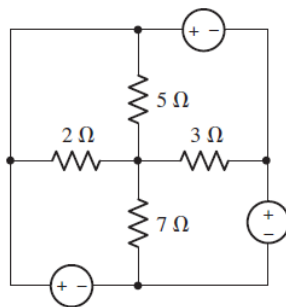


Figure 9: Prob.9

10. Define three clockwise mesh currents for the circuit of Fig. 10 , and employ mesh analysis to obtain a value for each.

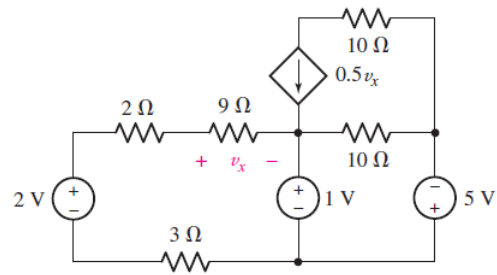


Figure 10: Prob.10

11. Employing the supermesh technique to best advantage, obtain numerical values for each of the mesh currents identified in the circuit depicted in Fig. 11.

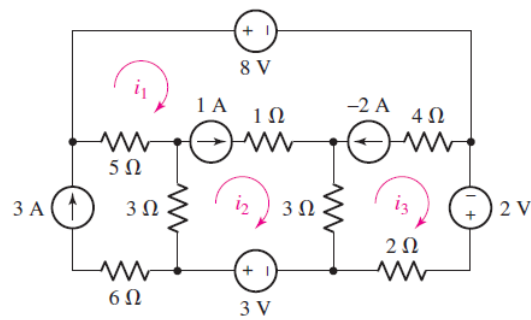


Figure 11: Prob.11

12. Determine the power absorbed by the $10\ \Omega$ resistor in Fig. 12.

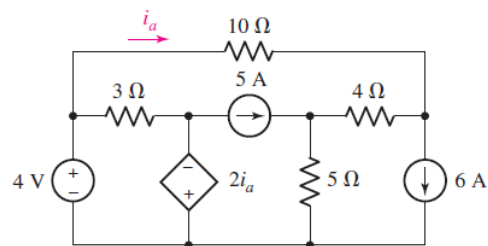


Figure 12: Prob.12